作业（共3页，第一页）

- 事件域：

1. 设 $\Omega = \bigcup_{i=1}^{n} A_i$，且事件 $A_i, i = 1, \ldots, n$ 互不相交，每个事件 A_i 发生的概率为正数。设 $\mathscr{F}$ 是包含所有 A_i 的最小事件域，问 $\mathscr{F}$ 中有多少个元素？

(选做) 1*.设 $\mathscr{F}_1, \mathscr{F}_2$ 都是 Ω 上的事件域，验证 $\mathscr{F}_1 \cap \mathscr{F}_2$ 也是 Ω 上的事件域。

- 概率的连续性：

2. 设 $\{A_n, n = 1, 2, \ldots\}$ 是 $\mathscr{F}$ 中的事件列，证明：如果 $\lim_{n\to\infty} A_n$ 存在，那么 $P(\lim_{n\to\infty} A_n) = \lim_{n\to\infty} P(A_n)$.

(选做) 2*. 选做、不计分、推荐 Lecture3 结束后再做) 证明 Borel-Cantelli 引理. (Hint: 证明 (2) 时，可利用不等式 $1-|x| \leq e^{-|x|}$)

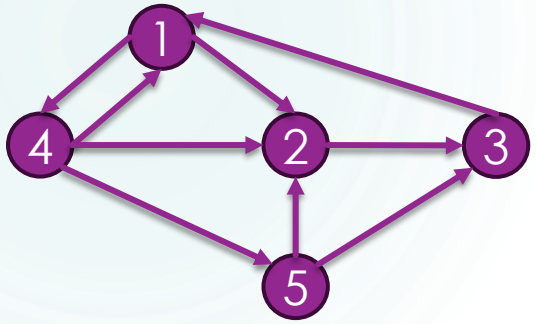
作业（共3页，第二页）

- 条件概率：
  3. 教材第一章 15;
  4. （Galton’s Paradox）投掷三枚质地均匀的硬币，至少两个硬币朝上的面是一样的，同时第三个硬币正与反是等概率的。因此三个硬币都是同一个面朝上的概率是$\frac{1}{2}$，你同意这个论断吗?
- 全概率：
  5. 某年级一共三个班级，一班3个男生、7个女生；二班7个男生、13个女生；三班12个男生、8个女生。新年时要抽取两位给幸运奖。先按人数比例，以1:2:2的概率选择其中一个班级，再从中先后选出两位同学。求：
     ① 先抽到男生的概率；
     ② 已知第二个抽到的是女生，求先抽到的是男生的概率。
  6. 教材第一章 22;
  7. 教材第一章 51;

作业（共3页，第三页）

- 全概率：

8. (**PageRank**) The World Wide Web is a collection of linked web pages. We can use a graph to represent these pages and their links: the nodes of the graph are pages $\mathcal{A}$ and there is a directed edge from i to j if page i has a link to j. We can define the rank of page i as $\pi(i) = \sum_{j\in\mathcal{A}} \pi(j)P(j,i)$, where $P(j,i)$ is the fraction of links in j that point to i and is zero if there is no such link. (*The basic idea of the algorithm is due to Larry Page, hence the name PageRank. Since it ranks pages, the name is doubly appropriate*). For convenience, we further restrict $\sum_{j\in\mathcal{A}} \pi(j) = 1$. Please solve $\pi(i), i = 1, \dots, 5$ in the following graph.



For example, we have $P(3,1) = 1, P(4,1) = \frac{1}{3}, P(5,1) = 0$, and $\pi(1) = \pi(2)\cdot 0 + \pi(3)\cdot 1 + \pi(4)\cdot\frac{1}{3} + \pi(5)\cdot 0$ in this graph.

具体题目

Problem 15. A coin is tossed twice. Alice claims that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Is she right? Does it make a difference if the coin is fair or unfair? How can we generalize Alice's reasoning?

Problem 22. Each of k jars contains m white and n black balls. A ball is randomly chosen from jar 1 and transferred to jar 2, then a ball is randomly chosen from jar 2 and transferred to jar 3, etc. Finally, a ball is randomly chosen from jar k. Show that the probability that the last ball is white is the same as the probability that the first ball is white, i.e., it is $m/(m+n)$.

Problem 51. An urn contains m red and n white balls.

(a) We draw two balls randomly and simultaneously. Describe the sample space and calculate the probability that the selected balls are of different color, by using two approaches: a counting approach based on the discrete uniform law, and a sequential approach based on the multiplication rule.

(b) We roll a fair 3-sided die whose faces are labeled 1,2,3, and if k comes up. we remove k balls from the urn at random and put them aside. Describe the sample space and calculate the probability that all of the balls drawn are red. using a divide-and-conquer approach and the total probability theorem.